2.2 Numbers

Informally we consider the set of integers

$$\mathbb{Z} = \{..., -3, -2, -1, 0, 1, 2, 3, ...\}$$

and the subset of natural numbers

$$\mathbb{N} = \{0, 1, 2, ..., \}$$

Although it's possible to construct the integers painstakingly from a small set of axioms (see ...) we instead for brevity simply state the most commonly used results as axioms.

2.2.1 Integers

We suppose the existence of a set $\mathbb{Z}$ with distinguished elements $0 \neq 1$ together with

- A binary operation

  $$+ : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$$

  and an involution

  $$(-) : \mathbb{Z} \to \mathbb{Z}$$

  satisfying

  - $-0 = 0$
  - $-(-x) = x$
  - $x = -x \iff x = 0$
  - $x + 0 = 0 = 0 + x$
  - $x + y = y + x$
  - $(x + y) + z = x + (y + z)$
  - $x + (-x) = 0 = (-x) + x$
  - $-(x + y) = (-x) + (-y)$

- A subset $\mathbb{N}$ such that

  - $0, 1 \in \mathbb{N}$
  - $x, y \in \mathbb{N} \implies x + y \in \mathbb{N}$
  - $x \in \mathbb{Z} \implies (x \in \mathbb{N}) \vee (-x \in \mathbb{N})$
  - $x \in \mathbb{N} \wedge -x \in \mathbb{N} \implies x = 0$

  which also satisfies the principle of induction

- Let $S \subseteq \mathbb{N}$ be a set such that
  - $0 \in S$
  - $x \in S \implies x + 1 \in S$

  then $S = \mathbb{N}$

It's possible to use these to show the existence of multiplication

Proposition 2.2.1 (Multiplication exists)
There exists a binary operation

$$\cdot : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$$

such that

- $x \cdot 0 = 0 = 0 \cdot x$
- $x \cdot 1 = x = 1 \cdot x$
- $xy = yx$
- $(xy)z = x(yz)$

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- $x(y+z) = xy + xz$
- $(y+z)x = yx + zx$
- $(-x)(y) = -(xy) = x(-y)$

We may also show the existence a partial ordering

Proposition 2.2.2 (Order exists)
There exists a relation $\leq$ on $\mathbb{Z}$ given by

$$x \leq y \iff y - x \in \mathbb{N}$$

which satisfies

$$x \leq y \vee y \leq x$$

$$x \leq y \wedge y \leq x \implies x = y$$

Define $x < y$ in the obvious way then it satisfies the usual trichotomy law, namely precisely one of the following holds

$$x < y, \quad x = y, \quad y < x$$

and further

- *$z > 0$ then $x < y \iff xz < yz$*
- *$z < 0$ then $x < y \iff yz < xz$*
- *$y > 1$ and $x > 0$ then $x < xy$*

Finally we can construct an absolute value function

Proposition 2.2.3
*There exists an **absolute value** function*

$$|\cdot| : \mathbb{Z} \to \mathbb{N}$$

such that

$$|x| = \begin{cases} x & 0 < x \\ 0 & x = 0 \\ -x & x < 0 \end{cases}$$

It satisfies

- $|x| = 0 \iff x = 0$
- $|x| = |y| \iff x = \pm y$
- $|xy| = |x||y|$
- $|x + y| \leq |x| + |y|$

In many cases it may be more convenient to use the following form of induction

Proposition 2.2.4 (Well-Ordering Principle)
Let $S \subset \mathbb{N}$ be a non-empty subset. Then it contains a minimal element d.

2.2.2 Arithmetic

Proposition 2.2.5 (Division Algorithm)
Let $x, y \in \mathbb{Z}$ be non-zero integers then there exists q, r such that

$$x = yq + r$$

and

$$0 \leq r < |y|$$

Furthermore (q, r) is the unique such pair.

Proof. Suppose first that $x, y > 0$. Let $S = \{x - yn \mid n \in \mathbb{Z}\} \cap \mathbb{N}$. Then $x \in S$ so it is non-empty. By the Well-Ordering principle it has a minimal element r. By assumption

$$x = yq + r$$

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for some $q \in \mathbb{Z}$ and $r \geq 0$. Suppose $r \geq y$, then $0 \leq x - y(q+1) < r$ contradicting minimality.

The case $x > 0, y < 0$ is then straightforward, as is the case $x < 0$.

For uniqueness suppose $yq' + r' = yq + r$ then $|y||q - q'| = |r' - r| < |y|$ from which it follows $|q - q'| = 0 \implies q = q' \implies r = r'$. ■

Corollary 2.2.6 (Ideals are Principal)
Let $S \subseteq \mathbb{Z}$ be a non-empty set such that

$$x, y \in S \implies x \pm y \in S$$

Then $S = d\mathbb{Z}$ for a unique $d \geq 0$.

Proof. First we claim that $0 \in S$. For if $x \in S$ then $0 = x - x \in S$ by assumption. Furthermore $x \in S \implies -x = 0 - x \in S$.

Consider the set $S' = (S \cap \mathbb{N}) \setminus \{0\}$. If it's empty then $S = \{0\}$ (for $x \in S \implies -x \in S$) and $d = 0$.

Otherwise it has a minimal element $d > 0$ by the well-ordering principle. Then by induction $d\mathbb{Z} \subseteq S$. Conversely suppose $y \in S$ then by the division algorithm $y = qd + r$ with $0 \leq r < d$. By assumption $r = y - qd \in S$ and by minimality must be equal to 0. Therefore $y \in d\mathbb{Z}$ and $d\mathbb{Z} = S$ as required. ■

Definition 2.2.7 (Divisibility)
Let $x, y \in \mathbb{Z}$ be two integers. We say that x divides y if there exists a such that $ax = y$. In this case we write

$$x \mid y$$

and

$$\frac{y}{x}$$

for the unique integer a such that $ax = y$.

Lemma 2.2.8
Let $x, y \in \mathbb{Z}$ be two integers then

$$x \mid y \implies |x| \leq |y|$$

In particular $x \mid y \wedge y \mid x \implies x = \pm y$.

Proposition 2.2.9 (Bezout's Theorem)
Let x, y be non-zero integers. Then there exists a unique positive integer d such that

- *d is a common divisor of x, y*
- *For any other common divisor e we have $e \mid d$.*

Further there exists integers a, b such that $ax + by = d$. We write this as (x, y).

Proof. Let $S = \{ax + by \mid a, b \in \mathbb{Z}\}$. Then by (2.2.6) we have $S = d\mathbb{Z}$ for a unique $d > 0$. As $x, y \in S$ by definition d is a common divisor, and by definition $d = d \cdot 1 = ax + by$ for some integers a, b. Suppose e is a common divisor then $d = ax + by = e(ap + bq)$ and $e \mid d$ as required.

Any two such common divisors have $d = \pm d'$ by the previous Lemma. Since they are positive and non-zero we have $d = d'$. ■

Proposition 2.2.10
Let a, x, y be non-zero integers then

$$|a|(x, y) = (ax, ay)$$

In particular

$$\left(\frac{x}{x, y}, \frac{y}{x, y}\right) = 1$$

Proof. This follows from the characterization of (x, y) as the minimal positive integer in the set $\{mx + ny\}$. ■

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2.2.3 Prime Factorization

Definition 2.2.11
Let $x \in \mathbb{Z}$ be a non-zero integer. We say that x

- *is a **unit** if it's equal to 1 or -1.*
- *is **prime** if it's not a unit and $x \mid p$ implies $x = \pm 1$ or $x = \pm p$*
- ***composite** otherwise*

Lemma 2.2.12
Let p be a positive prime and a a non-zero integer. Then precisely one of the following holds

- $(p, a) = 1$
- $(p, a) = p$ *and* $p \mid a$

Proof. Note that (p, a) is positive and divides both p and a so the result follows by definition of prime. ■

Proposition 2.2.13 (Euclid's Lemma)
Suppose $x \mid ab$ then $\frac{x}{x,a} \mid b$.

In particular if p is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. First suppose that $(x, a) = 1$. Then by assumption $zx = ab$ and by Bezout's Theorem $mx + na = 1$ for some integers m, n. Multiply by z to find that $abm + na = a(bm + n) = z$. Therefore $a(bm + n)x = ab$ and cancel a to find $x \mid b$ as required.

For the general case define $x' = x/(x, a)$ and $a' = a/(x, a)$. Then by (2.2.10) $(x', a') = 1$. Furthermore it's clear that $x' \mid a'b$ so we have $x' \mid b$ by the special case just proven.

Finally suppose $p \mid ab$. If $(p, a) = 1$ then $p \mid b$ by the first result. By (2.2.12) if this does not hold then $p \mid a$ as required.■

Using these results we may show that there exists a unique factorization into primes, unique up to multiplication by a unit.

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